Energy Conversion
Fall 2023

Important Concepts of AC Circuit

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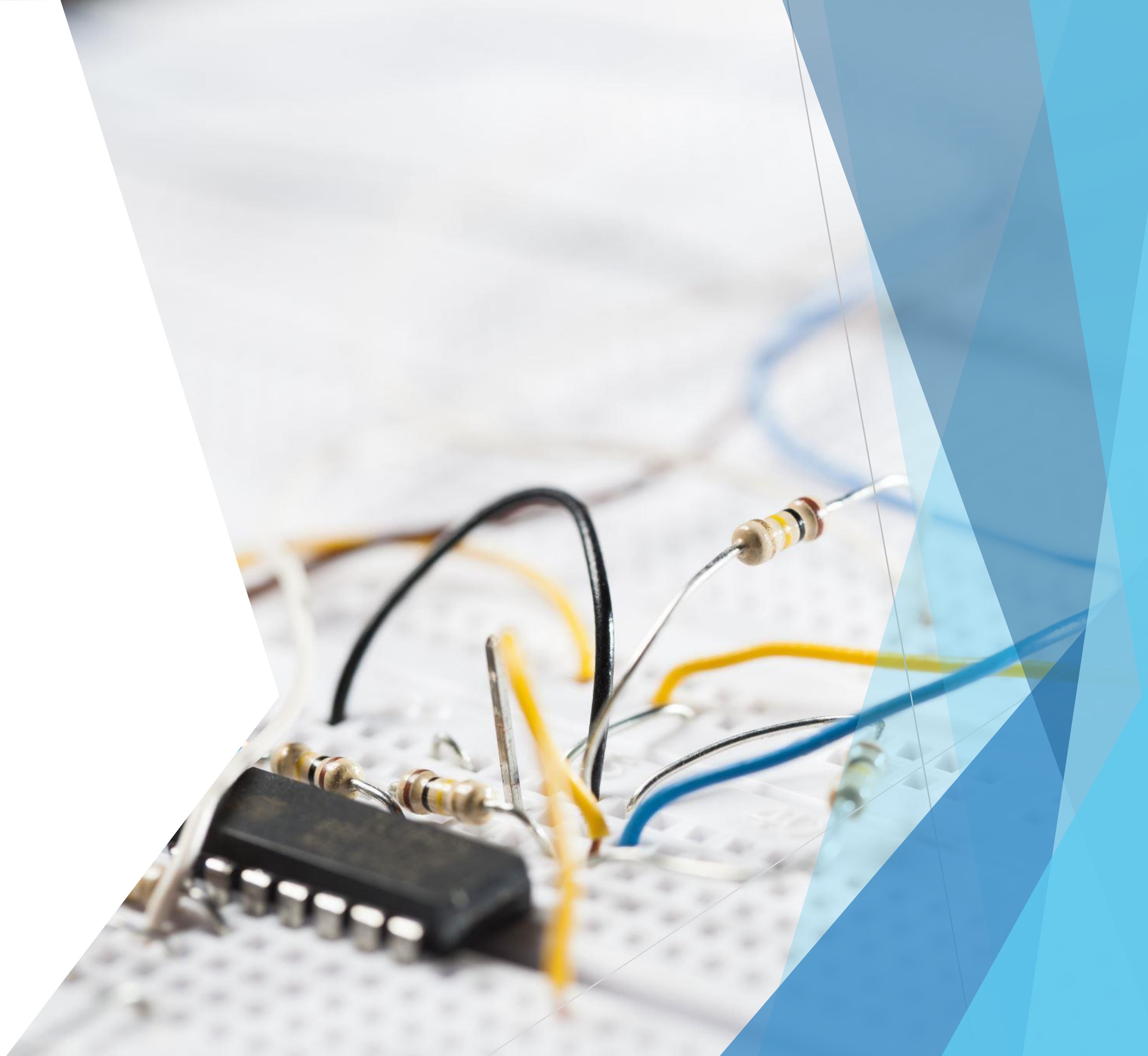
Summary

- Major variables in AC circuit: current (*i*), voltage (*v*), power (S, P, Q)
  - Parameters of the variables (amplitude, frequency, & phase)
  - Representations of the major variables
    - Complex number
    - Trigonometric functions
    - Phasor
  - Phase displacement
    - Where the phase displacement comes from
    - Impact of phase displacement
      - Impedance triangle
      - Lagging, leading, in-phase current with respective to the voltage
- Power
  - Instantaneous Power
  - Average (Active) and Reactive Power
  - Complex Power and Power triangle
  - Power factor
- Examples

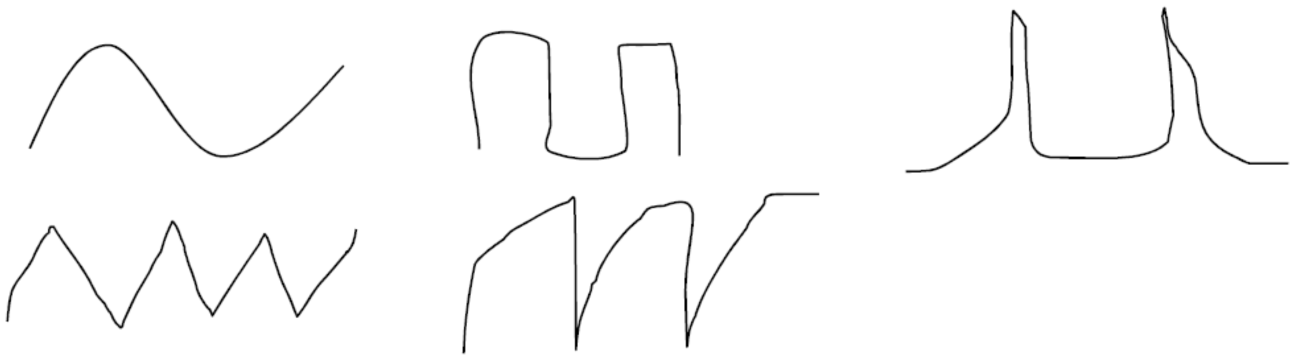
Important Concepts of AC Circuit



AC vs. DC Circuit

$V(t) = \text{constant},$
$I(t) = \text{constant}.$ → DC Electric circuit → DC Magnetic circuit

$V(t) \neq \text{constant},$
$I(t) \neq \text{constant}.$ → AC circuit → AC Magnetic circuit if not saturated



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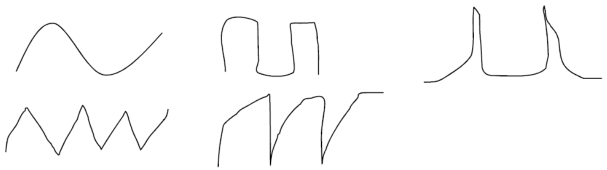
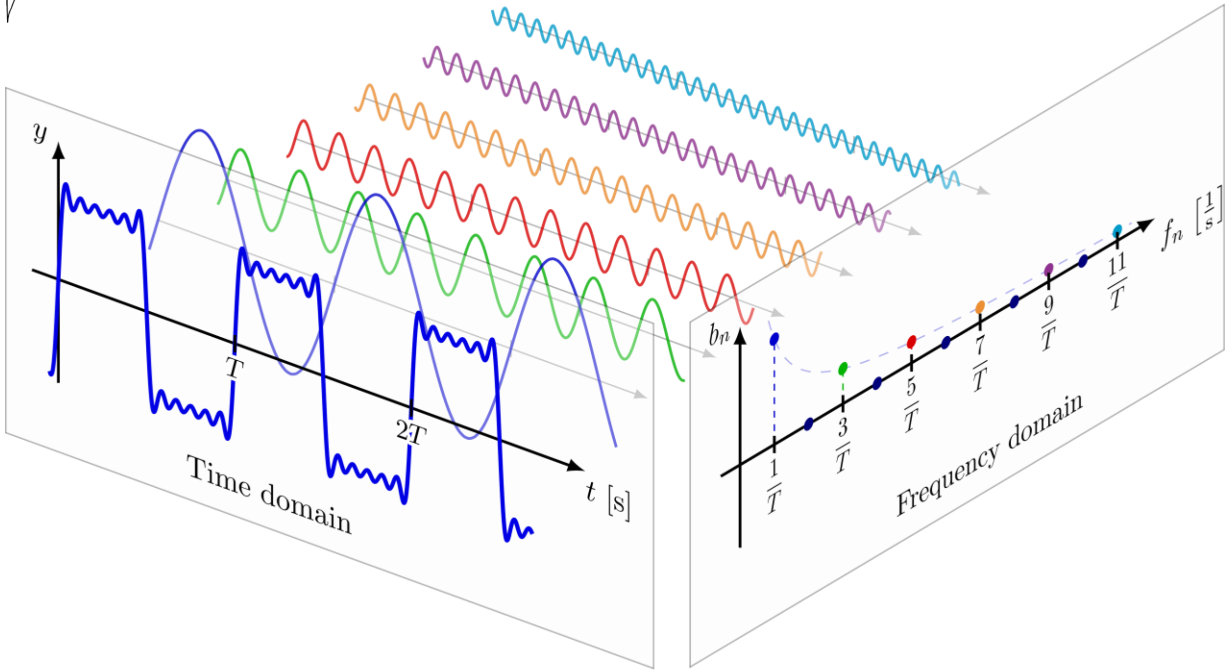




http://upload.wikimedia.org/wikipedia/commons/7/72/Fourier_transform_time_and_frequency_domains_%28small%29.gif
https://www.youtube.com/watch?v=k8FXF1KjzY0

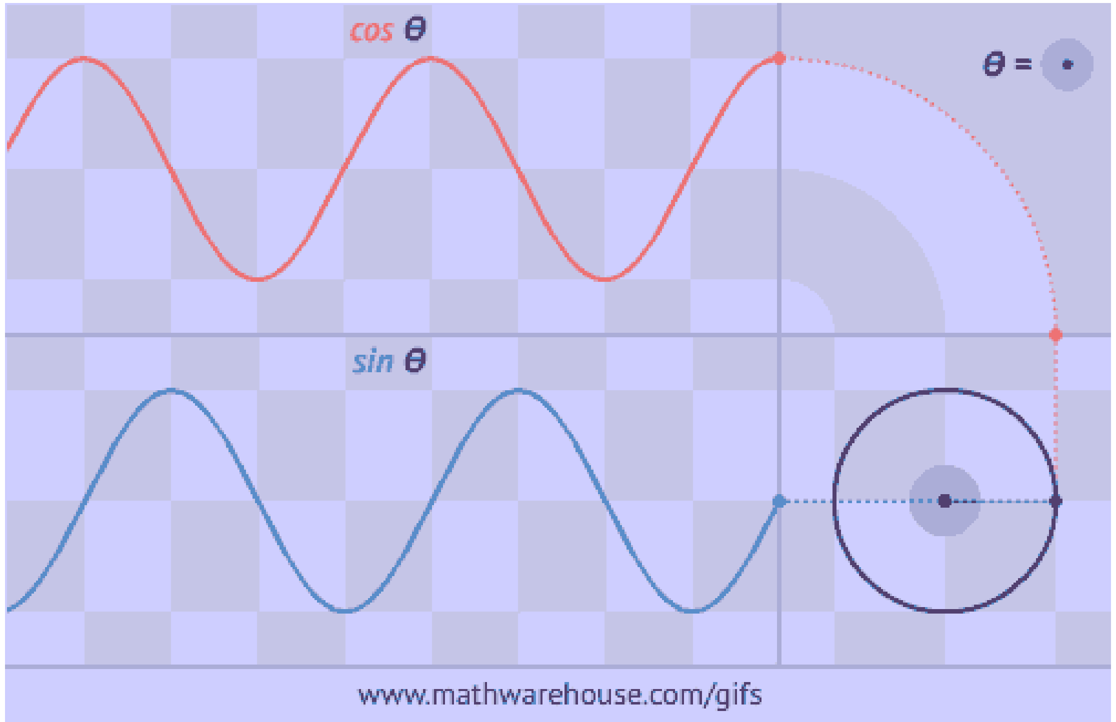
Phasor Circulation and Sine/Cosine Wave

- A phasor is a vector whose length is proportional to the maximum value of the variable.
- The phasor rotates counter-clockwise at the angular speed.

X axis

or

Y axis

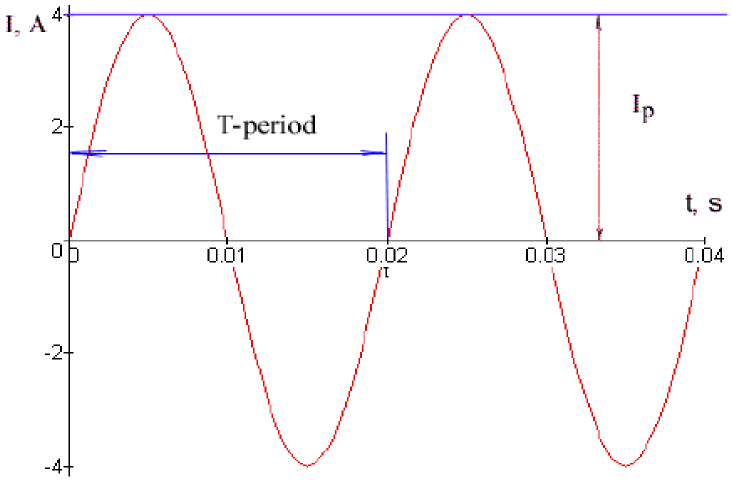


A Circulating vector
Phasor Diagram

Time or angle are the same if the angler speed of rotation is constant

https://www.youtube.com/watch?v=Q55T6LeTvsA
http://demonstrations.wolfram.com/PlottingASineOrCosineWave/

Description of Sine or Cosine Waves



$$I(t) = I_{\max}\sin(\omega t) = I_{\max}\sin(2\pi f t),$$
$$I(t) = \sqrt{2}I_{rms}\sin(\omega t) = \sqrt{2}I_{rms}\sin(2\pi f t)$$
$$\sqrt{2}I_{rms} = I_{\max}$$

$$\boxed{I_{\mathrm{rms}} = \sqrt{\frac{1}{T_2 - T_1}\int_{T_1}^{T_2} (I_{\mathrm{p}}\sin(\omega t))^2 dt}}$$

Frequency/Period
Phase angle and displacement
Amplitude/Magnitude: RMS, Max,

- AC (Alternating current):
- changing direction
  - Transient
  - Steady-state or stationary
  - ω=2p(60)=377 s-1 or rad/s
- AC circuit components
  - Resistor
  - Inductor (magnetic field)
  - Capacitor (electric field)
- Phasor relations
  - Phasor/phase
  - Leading, lag and in-phase

Review Basic Math Formulae

- **Integrals and derivatives of trig. functions:**

$$\frac{d\ \sin(\omega t)}{dt} = \omega \cos(\omega t)$$

$$\frac{d\ \cos(\omega t)}{dt} = -\omega \sin(\omega t)$$

$$\int \sin(\omega t)\,dt = -\frac{1}{\varpi}\cos(\omega t)$$

$$\int \cos(\omega t)\,dt = \frac{1}{\varpi}\sin(\omega t)$$

- **Trigonometric Identities**

$$\sin(\omega t + 90^{o}) = \cos(\omega t) \qquad \cos(\omega t + 90^{o}) = -\sin(\omega t)$$

$$\sin(\omega t - 90^{o}) = -\cos(\omega t) \qquad \cos(\omega t - 90^{o}) = \sin(\omega t)$$

$$\sin(x) \pm \sin(y) = 2\sin\left(\frac{x \pm y}{2}\right)\cos\left(\frac{x \mp y}{2}\right)$$

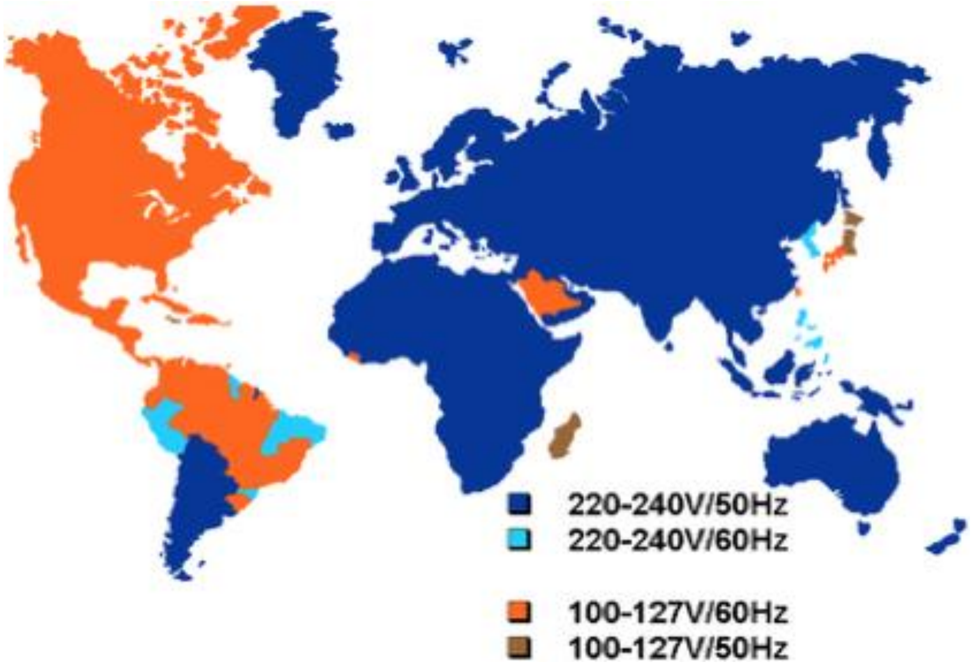
$$\cos(x) + \cos(y) = 2\cos\left(\frac{x + y}{2}\right)\cos\left(\frac{x - y}{2}\right)$$

$$\cos(x) - \cos(y) = -2\sin\left(\frac{x + y}{2}\right)\sin\left(\frac{x - y}{2}\right)$$

Complete the following identities by yourself: $\sin(x)\cos(y)$, $\sin(x)\sin(y)$ and $\cos(x)\sin(y)$

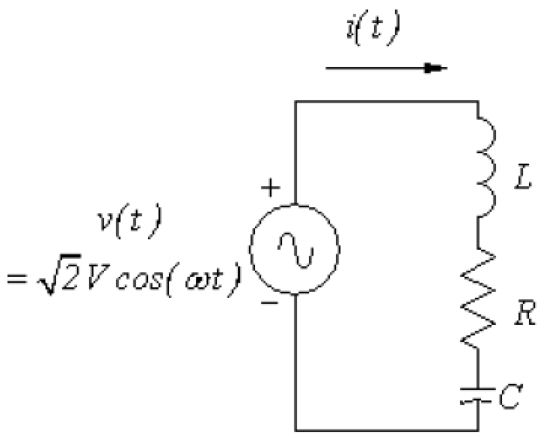
Frequency: 60 Hz vs 50 Hz

- ω is an angular frequency.
  - ω=2πf where f is the frequency in Hertz (Hz)
- In the US, the frequency is 60 Hz
  - ω=2π(60)=377 s^{-1} or rad/s
- In Europe and some countries in Asia, 50 Hz or 314 s^{-1}

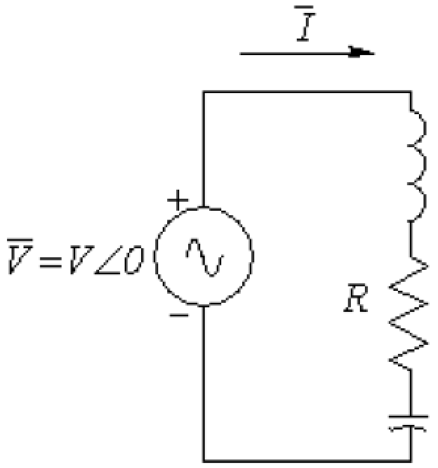


AC Circuit Structure

- **Calculate phasor domain impedance correspond to R, L, C.**

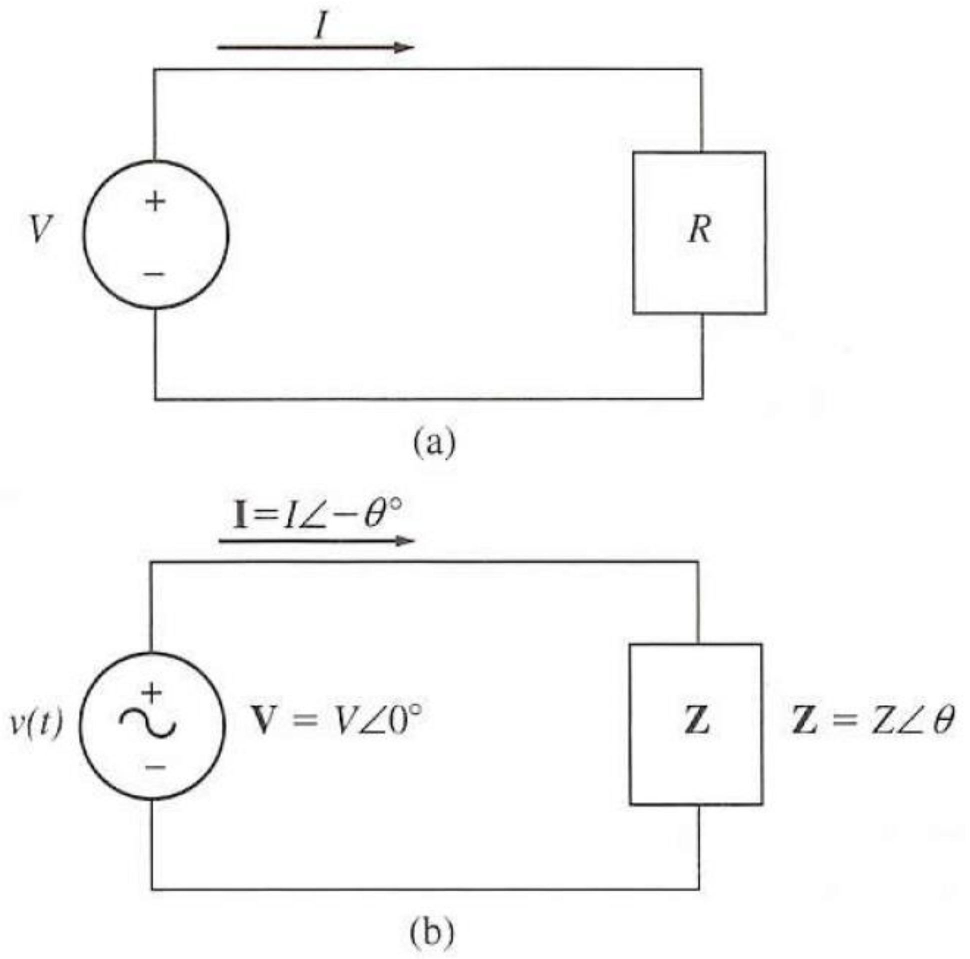


- **Find out the frequency, *f***
- **Find impedance angle**
- **Plot impedance triangle**

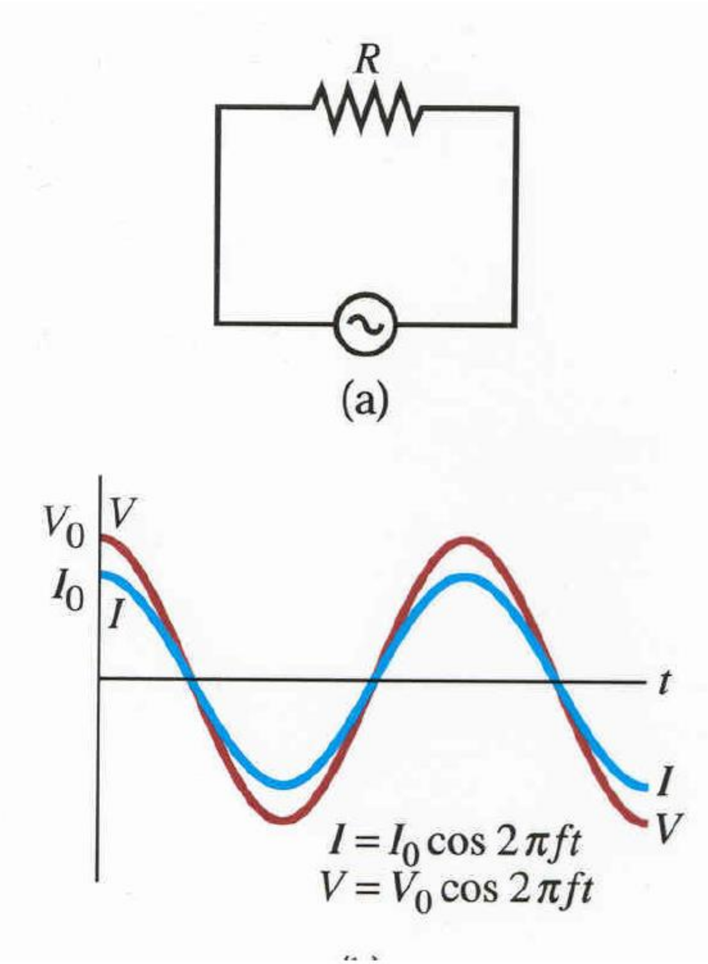


Phase: Phase Displacement of Voltage/Current

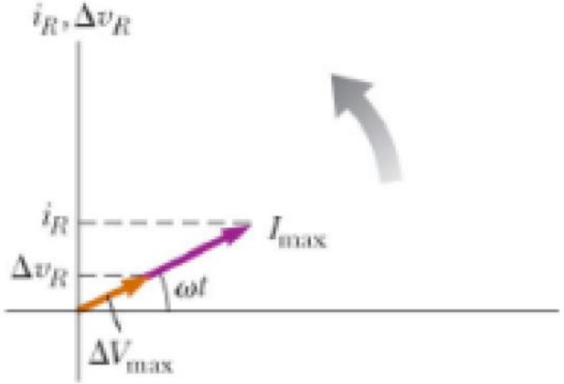
- **Voltage and Current**
- **In phase (pure resistor)**
- **Lagging current (positive impedance angle, inductive load)**
- **Leading current (negative impedance angle, capacitive load)**



In Phase

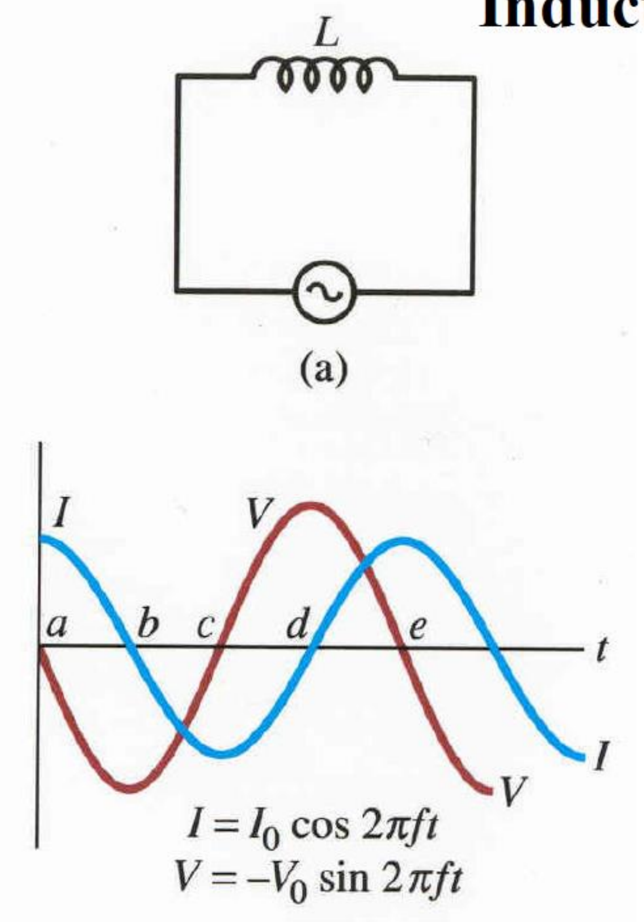


$$V_R = RI$$

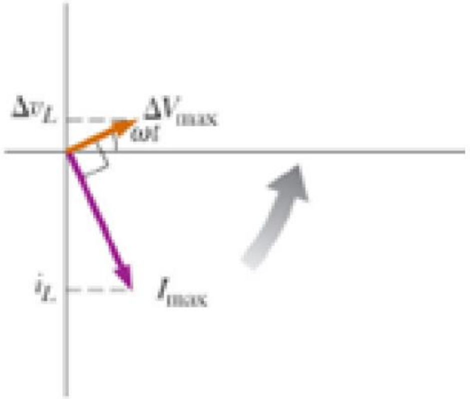


Lagging current

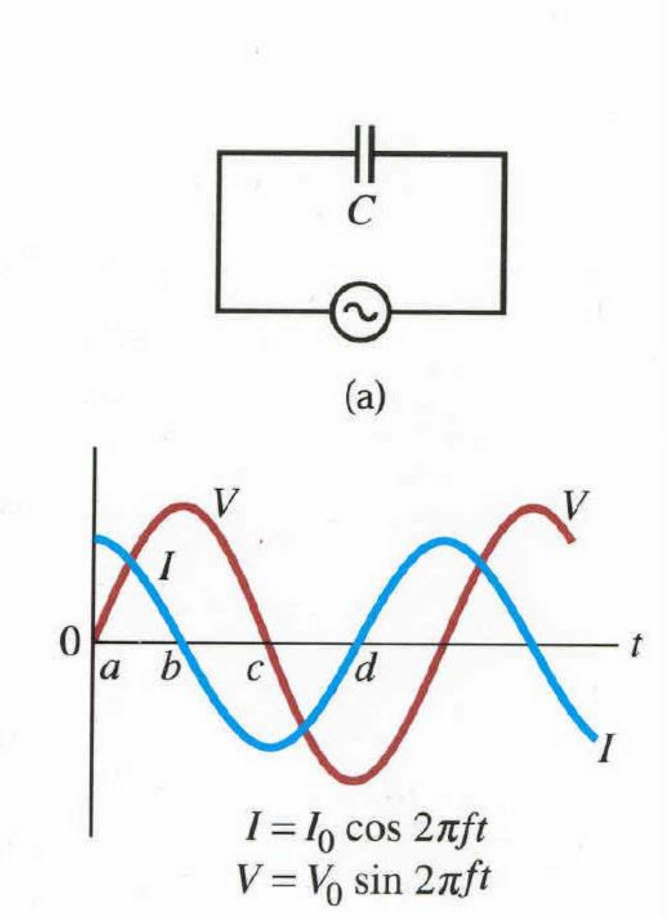
Inductor: current lags voltage



$$V_L = L\frac{dI}{dt}$$

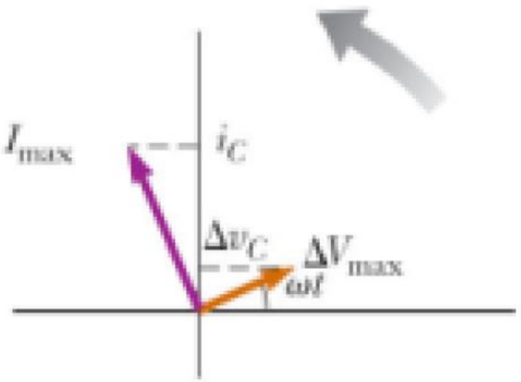


Leading Current



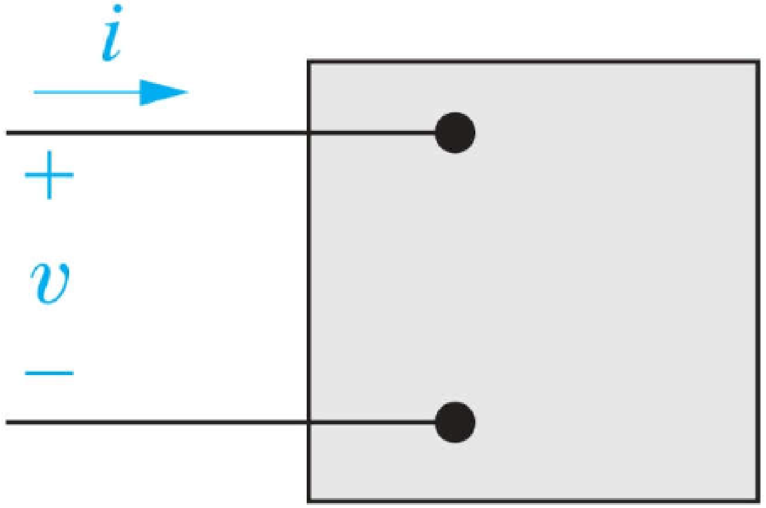
$$V_C = \frac{Q}{C} = \frac{1}{C}\int I dt$$

Capacitor: voltage lags current



Power

Instantaneous Power

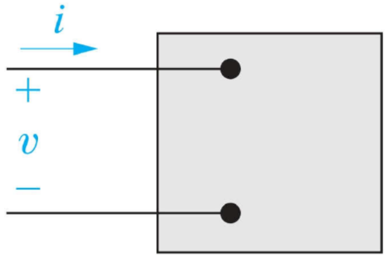


$$p = vi$$

$$v = V_m \cos(\omega t + \theta_v)$$

$$i = I_m \cos(\omega t + \theta_i)$$

Instantaneous Power



$$p = vi$$

$$v = V_m \cos(\omega t + \theta_v)$$

$$i = I_m \cos(\omega t + \theta_i)$$

$$p = V_m \cos(\omega t + \theta_v - \theta_i) I_m \cos \omega t$$

$$\cos\alpha \cos\beta = \frac{1}{2}\cos(\alpha - \beta) + \frac{1}{2}\cos(\alpha + \beta)$$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega t + \theta_v - \theta_i)$$

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

p is at twice the frequency of v and i

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\cos 2\omega t - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\sin 2\omega t$$

Average and Reactive Power

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\cos 2\omega t - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\sin 2\omega t$$

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) \longleftarrow$$ Average, or Real Power

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i) \longleftarrow$$ Reactive Power

$$p = P + P\cos 2\omega t - Q\sin 2\omega t$$

Real, Reactive, and Apparent Power in AC Circuits

- **In a DC circuit, supplied power to the load is P=VI.**
- **In an AC circuit, the instantaneous power supplied to the**
- **load is given by…**

$$p(t) = v(t)i(t) = 2VI\cos\omega t\cos(\omega t - \theta)$$

$$= \underbrace{VI\cos\theta(1+\cos 2\omega t)}_{VI\cos\theta + "0"} + \underbrace{VI\sin\theta\sin 2\omega t}_{"0"}$$

Real, Reactive, and Apparent Power in AC Circuits

- **Active or Real Power :**

$$P = V_{rms} I_{rms} \cos\theta = I_{rms}^2 Z \cos\theta = I_{rms}^2 R$$

- **Reactive Power :**

$$Q = V_{rms} I_{rms} \sin\theta = I_{rms}^2 Z \sin\theta = I_{rms}^2 X$$

- **Apparent Power :**

$$S = V_{rms} I_{rms}^{conj} = I_{rms}^2 Z = \sqrt{P^2 + Q^2}\,(scalar)$$

- **Complex Power :**

$$S = V_{rms} I_{rms}^{*} = |V_{rms}||I_{rms}|\cos\theta + j|V_{rms}||I_{rms}|\sin\theta = P + jQ$$

where θ is impedance angle from…

$$Z = Z\angle\theta = Z\cos\theta + jZ\sin\theta = R + jX$$

Complex Power and Power Factor

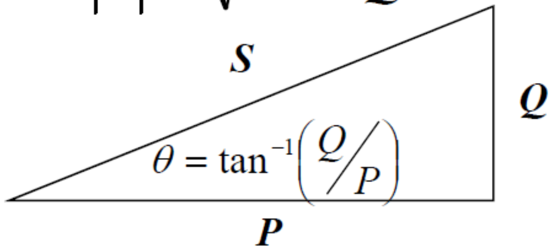
$$S = |S|(\cos\theta + j\sin\theta) = P + jQ$$

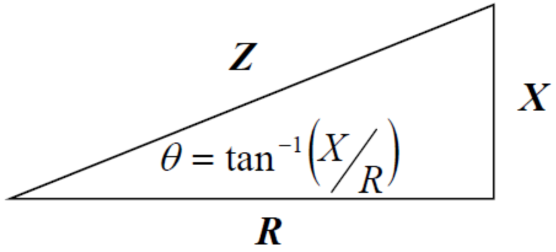
Complex power:
VA, kVA, MVA

Active power:
W, kW, MW

Reactive power:
Var, kVar, MVar

$$|S| = \sqrt{P^2 + Q^2}$$





Lagging current:
$Q > 0,\ \theta > 0$, inductive
Leading current:
$Q < 0,\ \theta < 0$, capacitive

$$P = |S|\cos\theta$$

$$Q = |S|\sin\theta$$

$\boxed{\text{Power factor}} = \cos\theta = P/|S|$

P, Q* > 0:
consuming/absorbed

P, Q* < 0:
producing/supplied

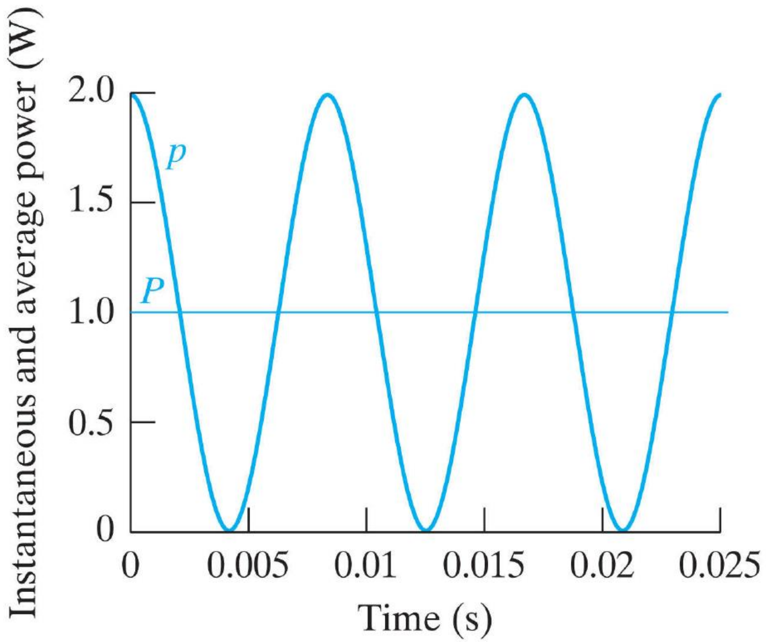
Power for Purely Resistive Circuits

- **For a purely resistive circuit, the voltage and current are in phase.**

$$\theta_v = \theta_i$$

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) = \frac{V_m I_m}{2}$$

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i) = 0$$



$$\boxed{p = P + P\cos 2\omega t - Q\sin 2\omega t \rightarrow P + P\cos 2\omega t}$$

Power for Purely Inductive Circuits

- **Voltage and Current are out of phase by 90°.**
- **The current lags the voltage by 90°.**

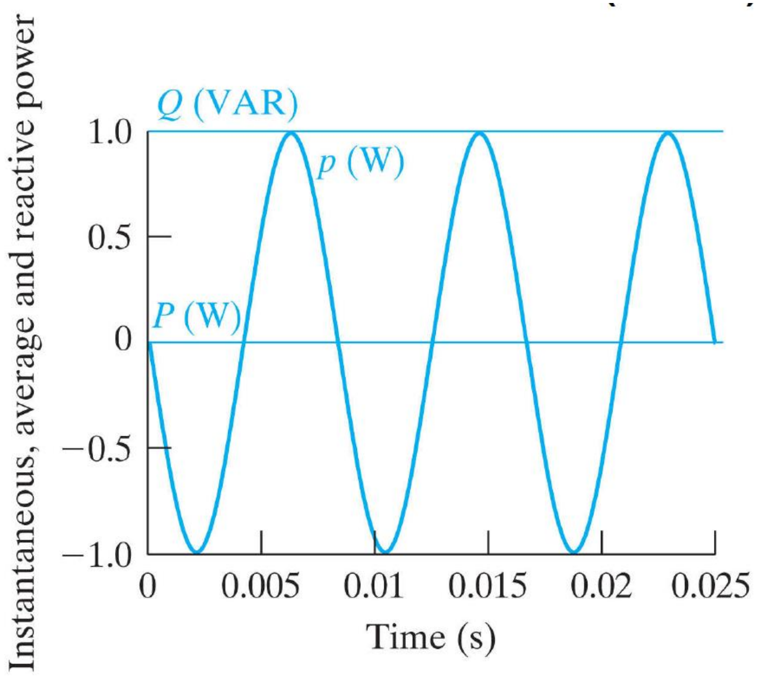
$$\theta_i = \theta_v - 90°$$

$$\theta_v - \theta_i = +90°$$

$$P = \frac{V_m I_m}{2}\cos(90°) = 0$$

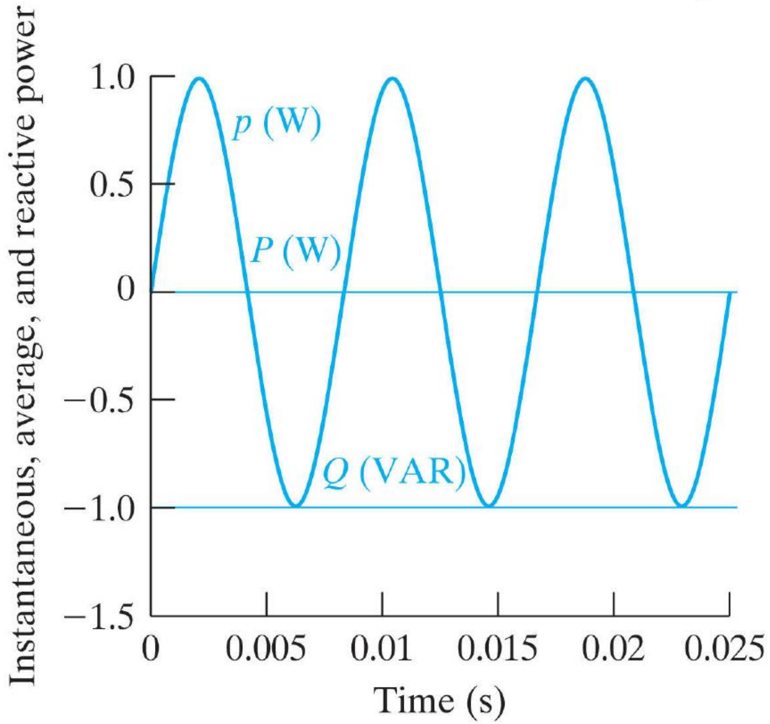
$$Q = \frac{V_m I_m}{2}\sin(90°) = \frac{V_m I_m}{2}$$

$$\boxed{p = P + P\cos 2\omega t - Q\sin 2\omega t \rightarrow -Q\sin 2\omega t}$$



Power for Purely Capacitive Circuits

- **Voltage and Current out of phase by 90°**
- **The current leads the voltage by 90°.**



$$\theta_i = \theta_v + 90°$$

$$\theta_v - \theta_i = -90°$$

$$P = \frac{V_m I_m}{2}\cos(-90°) = 0$$

$$Q = \frac{V_m I_m}{2}\sin(-90°) = -\frac{V_m I_m}{2}$$

$$\boxed{p = P + P\cos 2\omega t - Q\sin 2\omega t \rightarrow -Q\sin 2\omega t}$$

Power Factor

- Power factor angle

$$\theta_v - \theta_i$$

- Power factor

$$pf = \cos(\theta_v - \theta_i)$$

- Reactive factor

$$rf = \sin(\theta_v - \theta_i)$$

- Lagging power factor
  - Current Lags the Voltage
  - Inductive Load
- Leading power factor
  - Current Leads the Voltage
  - Capacitive Load

Questions